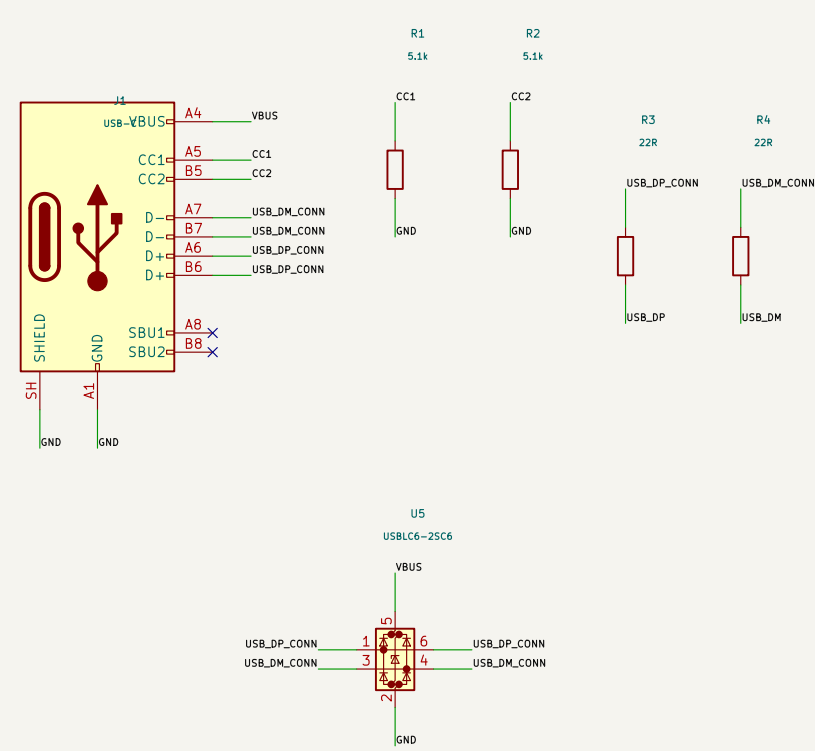
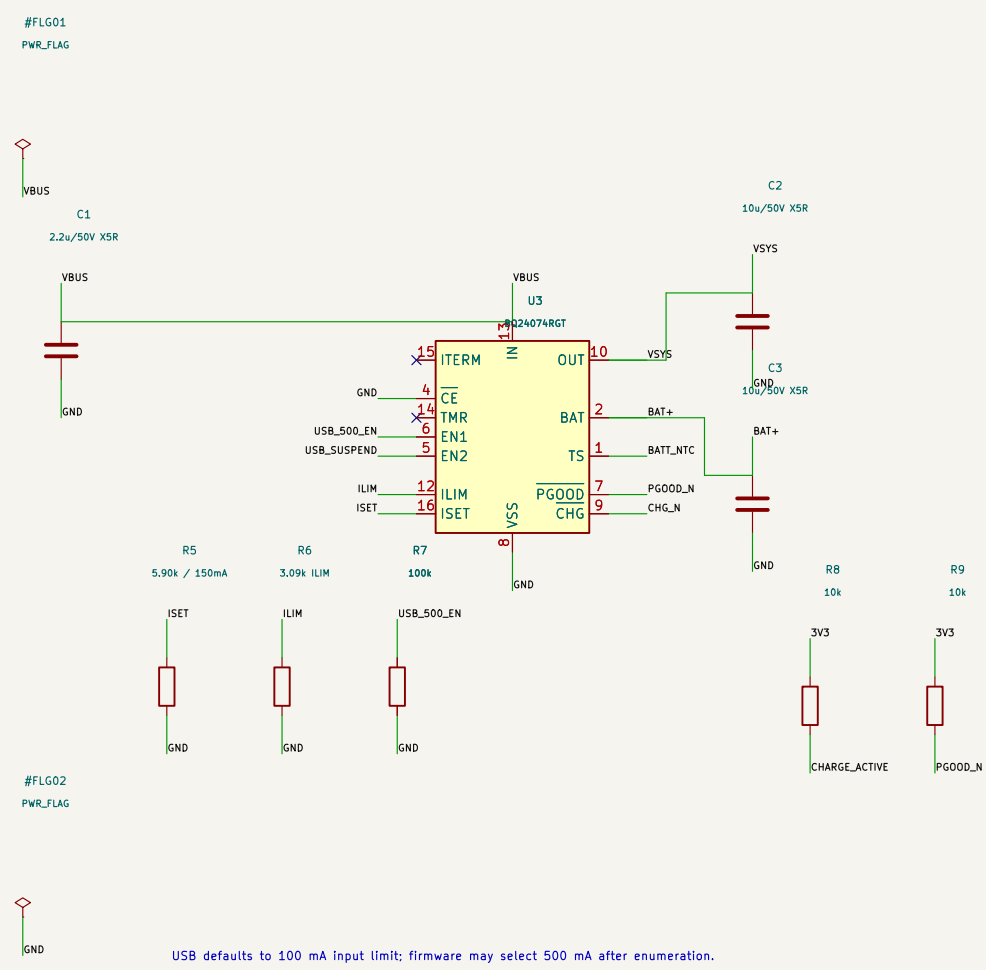


USB-C / 5 V INPUT / CC / DATA

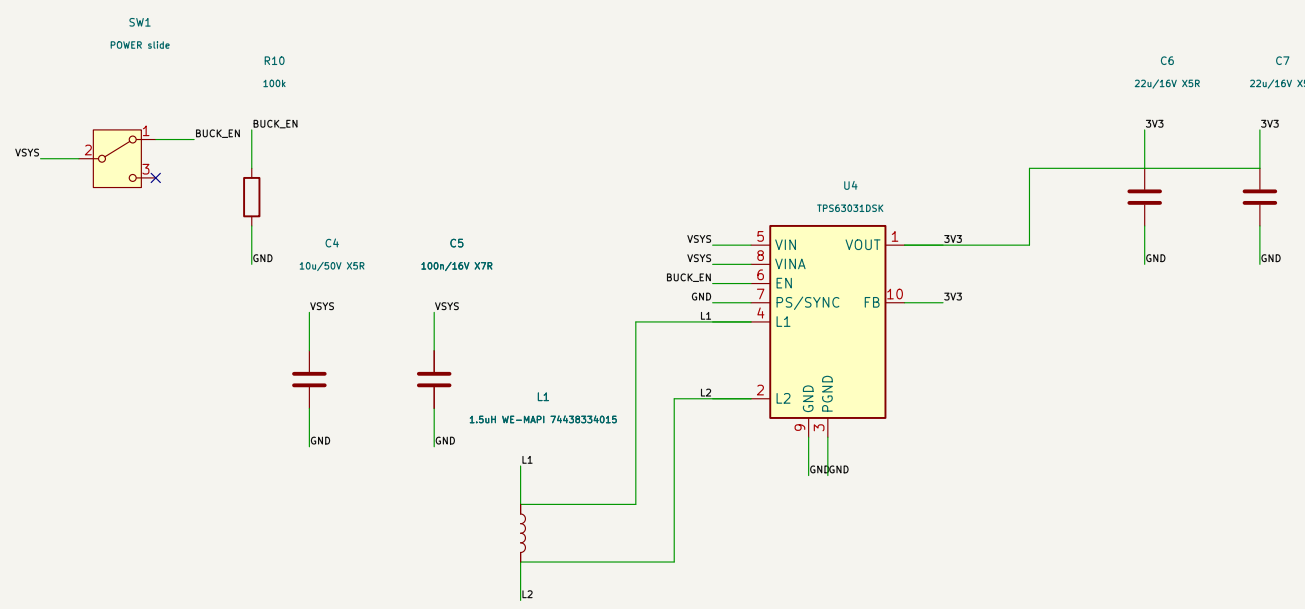


1S LI-ION CHARGE + POWER PATH

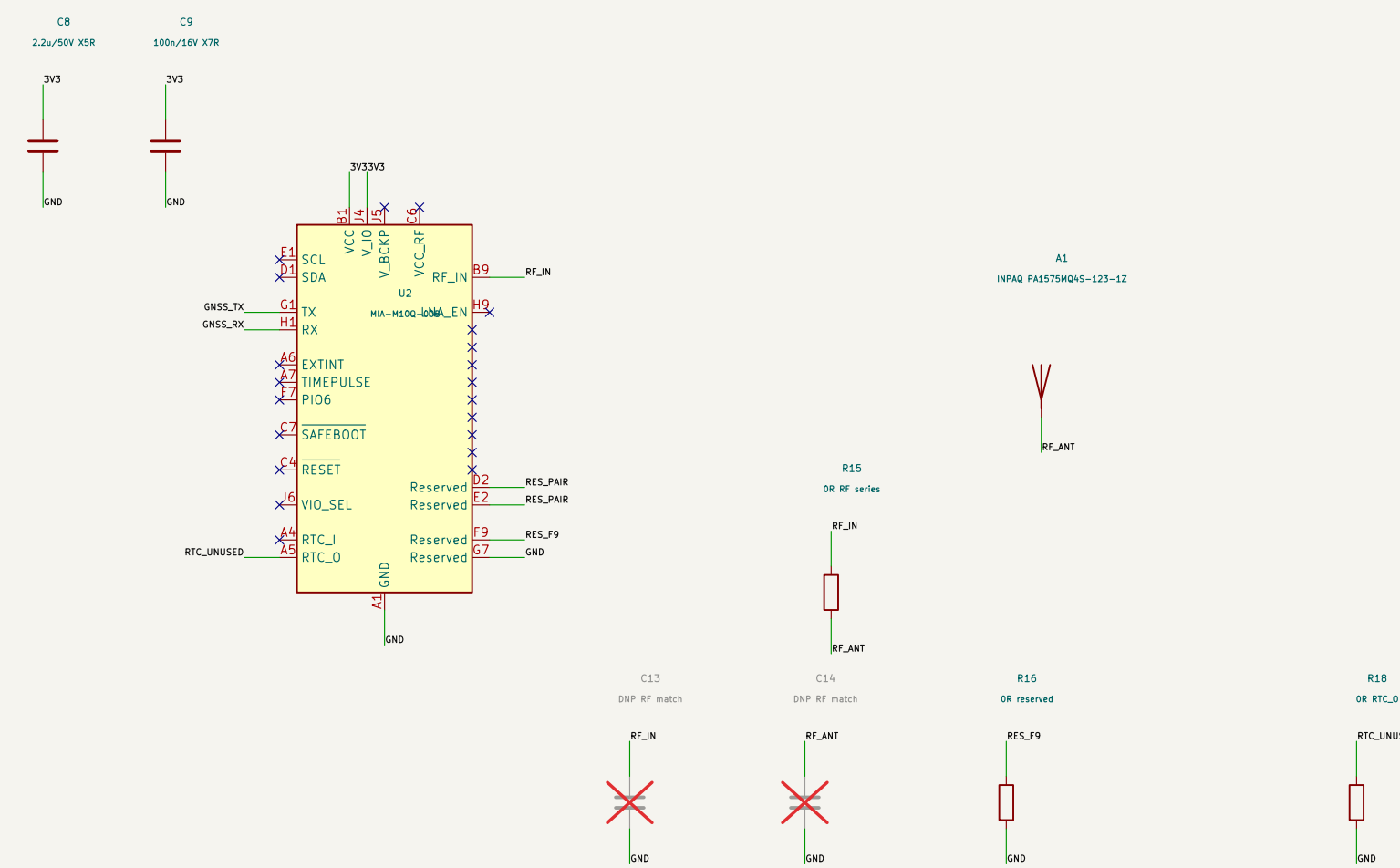


USB defaults to 100 mA input limit; firmware may select 500 mA after enumeration

3.3 V BUCK-BOOST

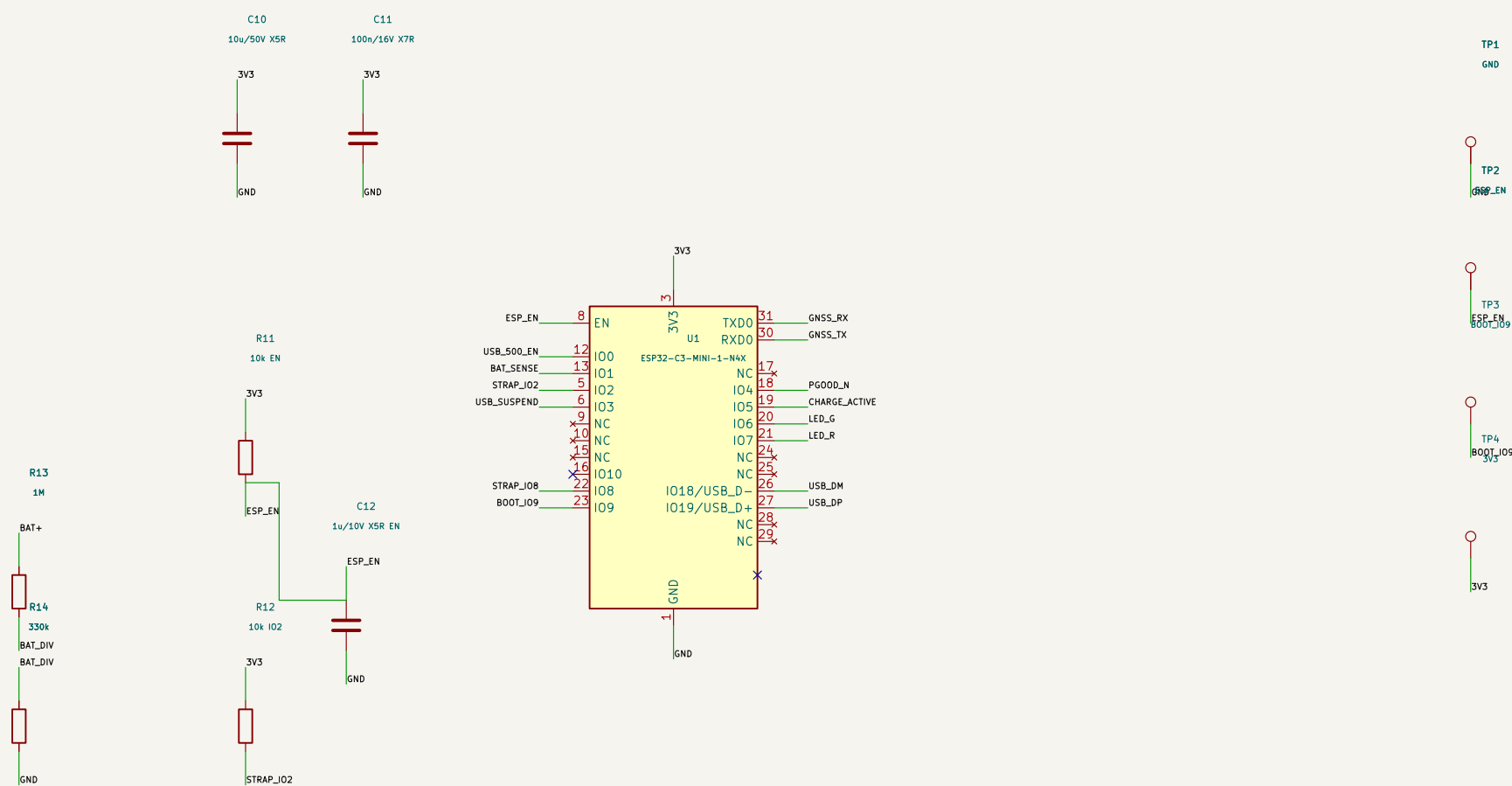


MIA GNSS + DIRECT CERAMIC PATCH / RF MATCH



INPAQ 10x10x4mm GPS L1 patch; engineered through-pin mount. Final Rf tuning and exact-part sourcing required

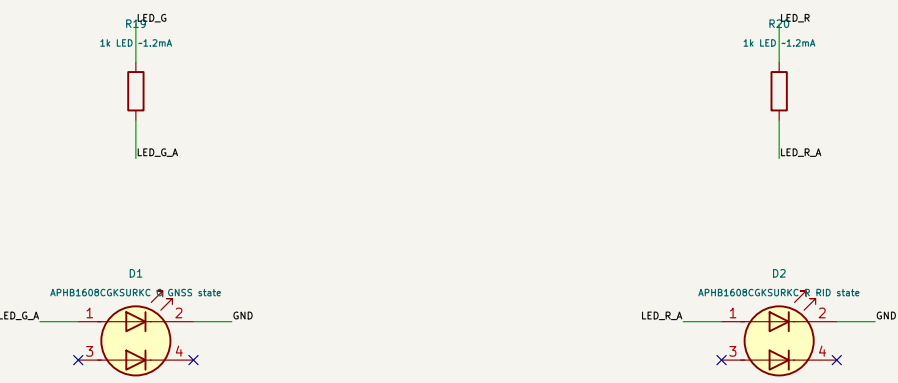
ESP32-C3 BLE/WI-FI + PROGRAMMING



ESP antenna is inside overall outline with host-board corner recess and all-layer keepout

Protected 1S pack with 10k NTC required; charge -150 mA is provisional.

STATUS INDICATORS / ISOLATED CHARGE SENSE



G: GP106, blisk acquiring / steady valid GNSS. R: GP107, actual successful RID transmit state.

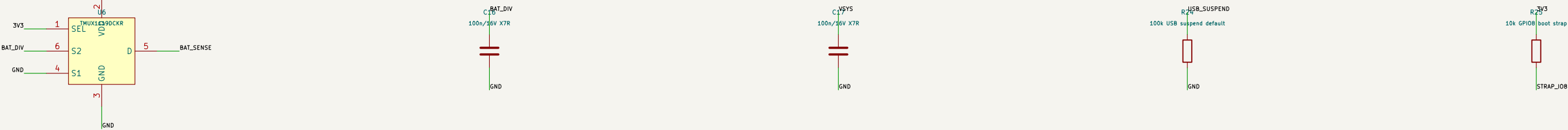
S green: 3V3 system rail enabled (not firmware-health proof). S red: charger CHG_N low, independent of MCU power. Charging can show amber while running.

Q1 prevents the VBUS-powered CHG node from driving an unpowered ESP input. Charging = (PGOOD_N low) AND (CHARGE_ACTIVE high); ignore CHARGE_ACTIVE without valid USB input.

G/R Indicators require future firmware. An Illuminated R is not proof of compliant RF/RID transmission.

POWER-OFF ADC ISOLATION / USB POLICY / RECOVERY STRAT

Native CAD checks passed; battery, sourcing, assembly and physical qualification pending.



U6: SEL=3Y3, Off -> ADC grounded; on -> BAT_DIV, C16 stays on divider side. Verify rail-collapse sequencing.

USB EN2/EN1: 00=100mA, 01=500mA, 11=suspend. Enter EN1 first; resume EN2 first. Avoid 10 ILIM state.

Default 100mA. S0F/connected is NOT a configured 500mA grant. Validate host grant before selecting USB500.

MCU-off charging cannot detect host suspend; USB power policy and prototype measurements remain release gates

GPIOB pull-up supports manual download with GPIO9/BOOT low. No status LED on strapping pins.

Chapter 182: GRIO3. Screenshot shows enabled system call: G/R use GRIO6/GRIO7 firmware state